

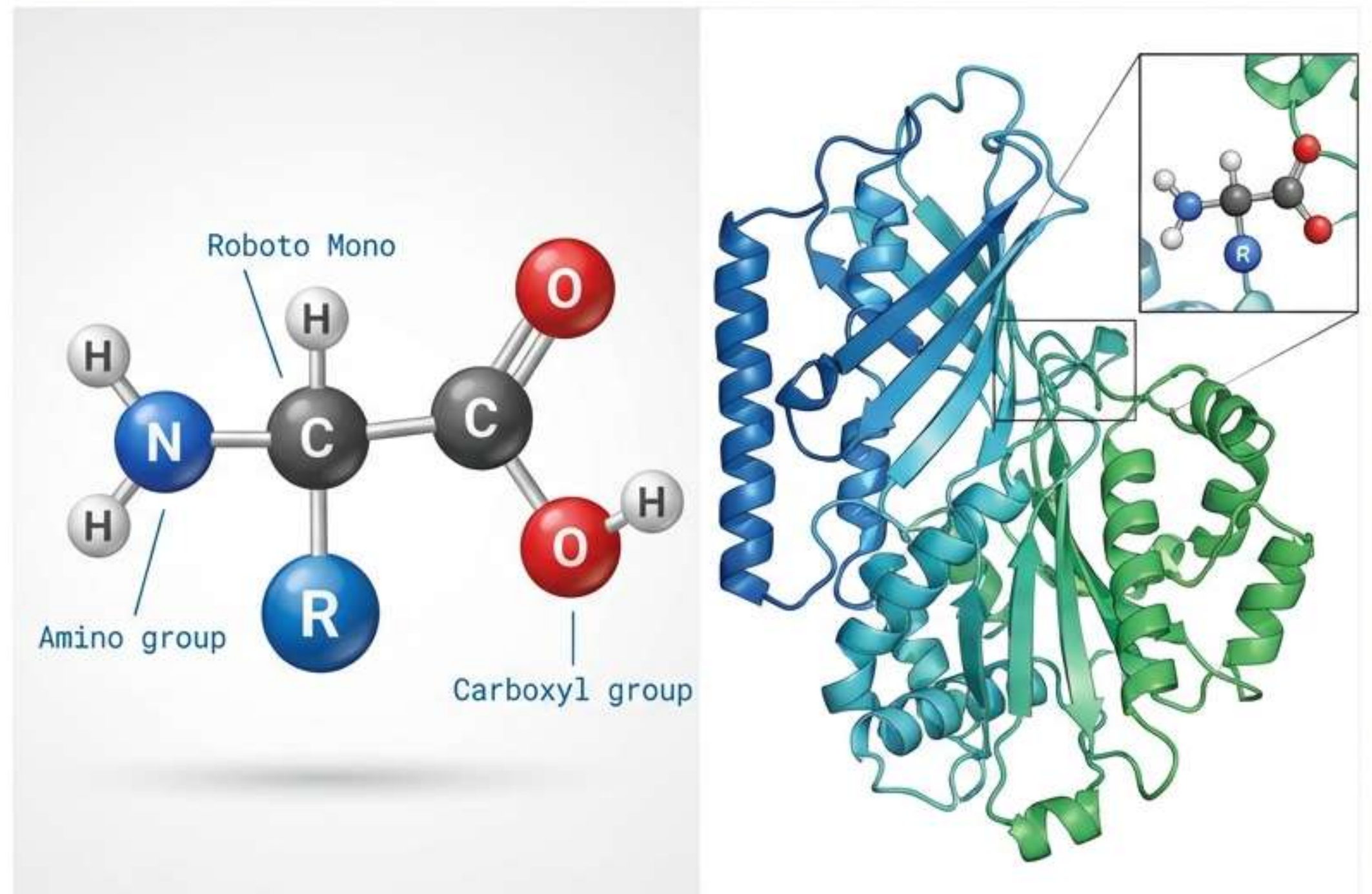
From Building Blocks to Biological Machines

Amino Acids & Proteins: A Comprehensive Study Guide for Dental Medicine

Core Definition: Amino acids are the structural units (monomers) of proteins, constituting >50% of the dry weight of cells.

The Narrative Arc:

1. **The Bricks:** Structure & Classification
2. **The Mortar:** Physical & Chemical Properties (pHi)
3. **The Architecture:** Protein Structure (1^o to 4^o)
4. **The Toolbox:** Purification & Analysis
5. **Exam Strategy:** Analysis of Past Questions & Traps



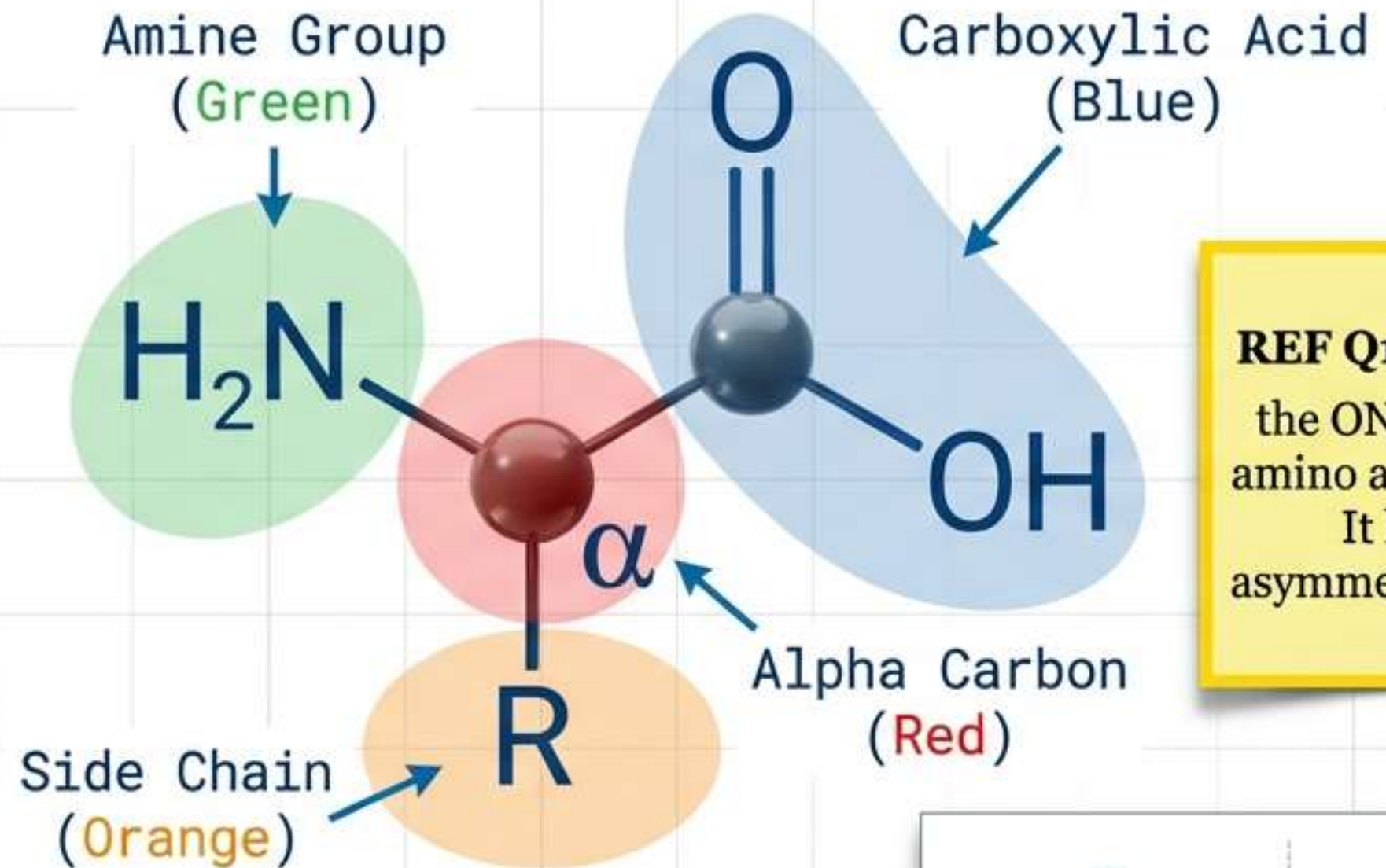
The Fundamental Structure: The Central Dogma

in Helvetica Now Display Bold in Scientific Blue

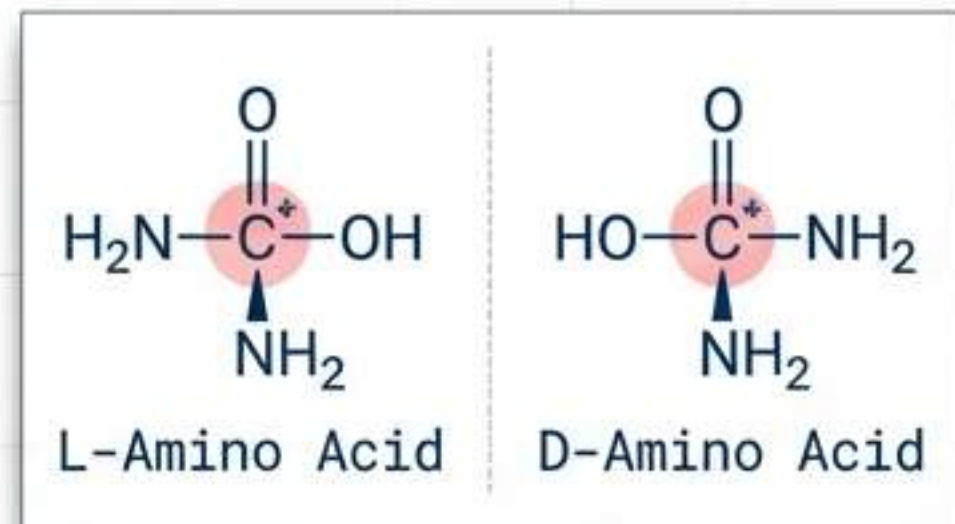
Common Structure: All AAs possess a carboxylic acid ($-\text{COOH}$) and a primary amine ($-\text{NH}_2$) on the same alpha-carbon (C_α).

Stereoisomerism:

- All AAs (Except **Glycine**) have an asymmetric carbon (C^*) and are **Chiral**.
- **Enantiomers:** D- and L-series. Natural proteins primarily use **L-amino acids**.



REF Q1: Glycine is the ONLY achiral amino acid ($\text{R} = \text{H}$). It has no asymmetric carbon.



Classification I: Polarity Overview

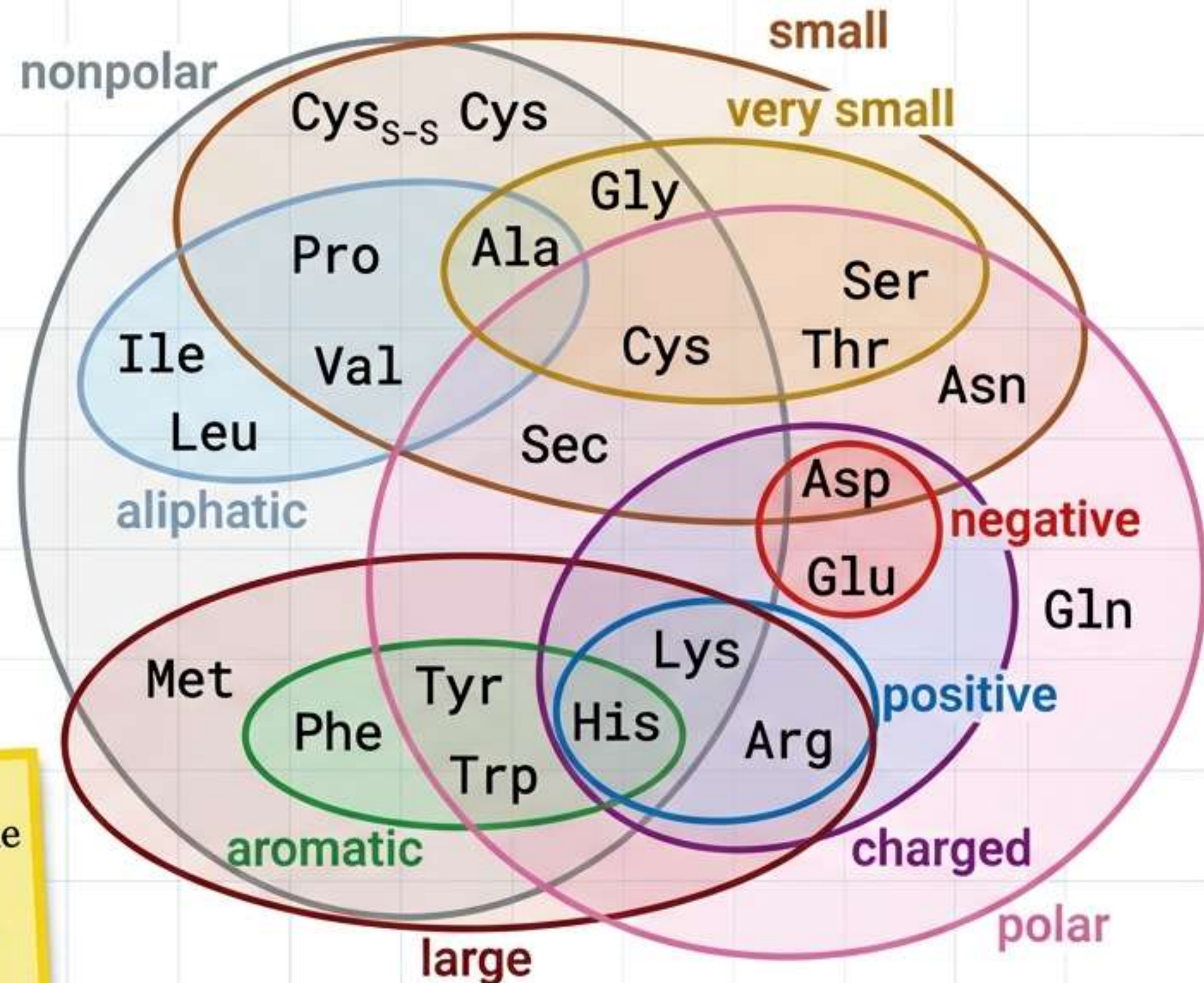
1. Non-Polar (Hydrophobic):

- Found in the *interior* of proteins (Hydrophobic Effect).
- Aliphatic: Gly, Ala, Val, Leu, Ile, Pro.
- Aromatic: Phe, Trp.
- Sulfur: Met.

2. Polar (Hydrophilic):

- **Uncharged:** Ser, Thr, Cysteine, Asn, Gln, Tyr.
- **Charged:** Asp, Glu (Negative) / Lys, Arg, His (Positive).

REF Q6: Leucine is Aliphatic, Branched, and Hydrophobic.



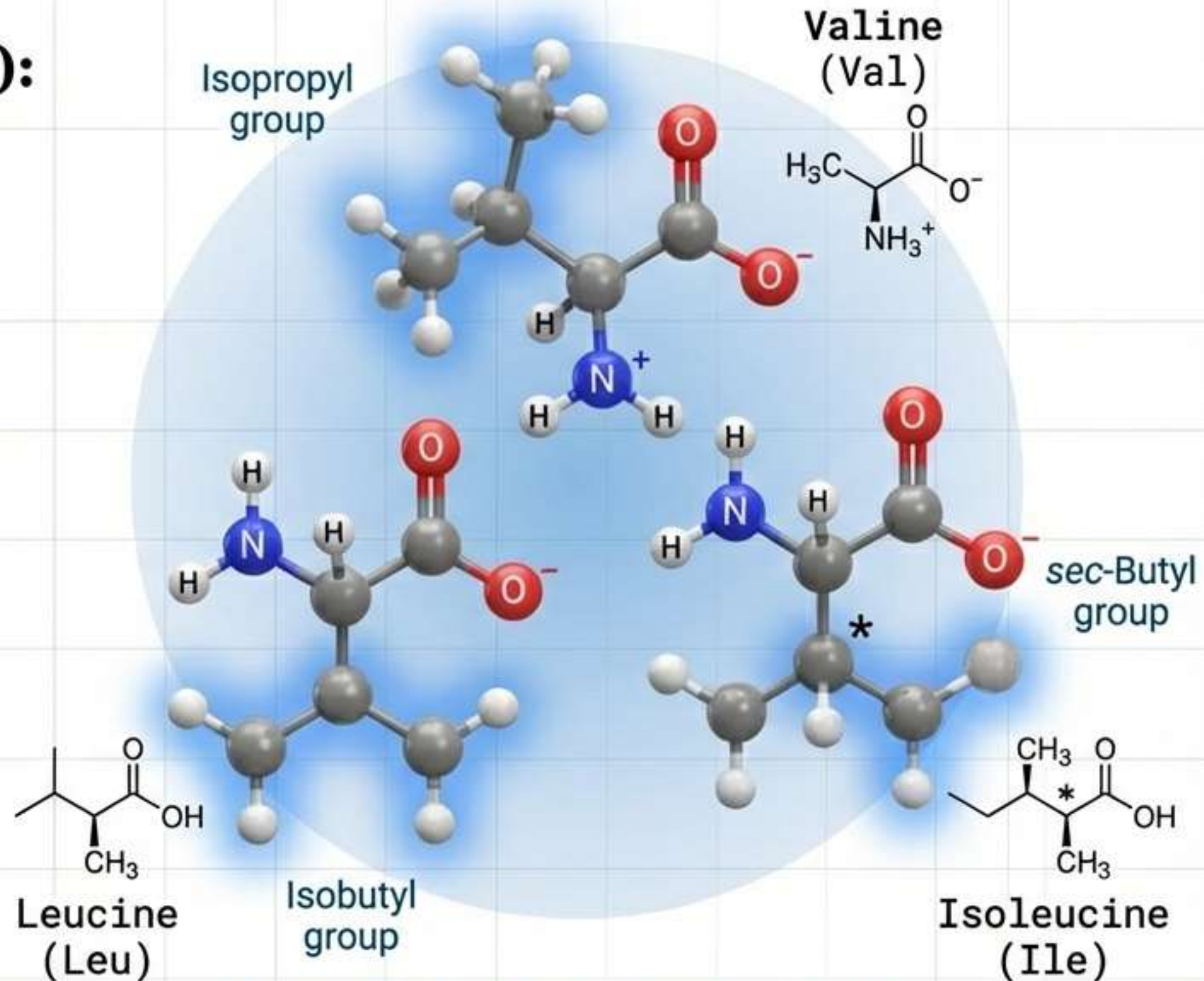
Classification II: Aliphatic & Branched Chain (BCAA)

Branched-Chain Amino Acids (BCAA):

- **Valine (Val, V):** Isopropyl group. Essential.
- **Leucine (Leu, L):** Isobutyl group. Essential & Strictly Ketogenic.
- **Isoleucine (Ile, I):** *sec*-Butyl group. Essential, Glucogenic & Ketogenic.

REF Q23: Valine is essential and non-polar.

REF Q3: Isoleucine (and Threonine) has TWO asymmetric carbons.

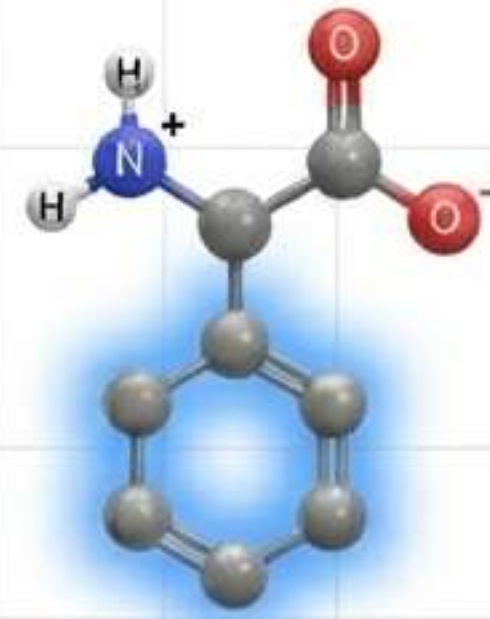


Branched-Chain Amino Acids

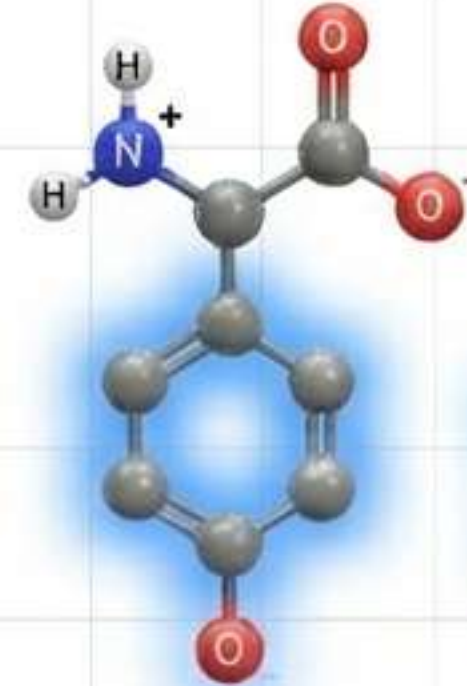
Classification II: Aromatic & Cyclic Side Chains

Aromatic Amino Acids:

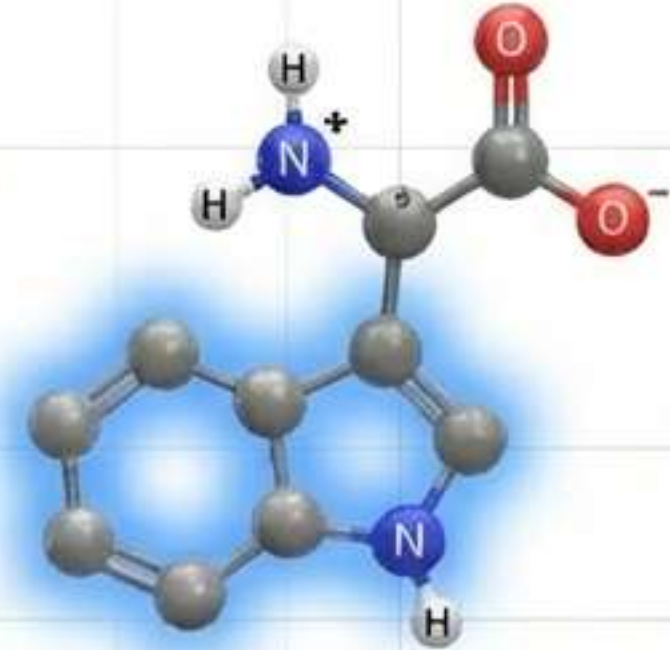
- **Phenylalanine (Phe, F):** Benzene ring. Hydrophobic.
- **Tyrosine (Tyr, Y):** Phenol group (-OH). Can be phosphorylated. Absorbance at 280nm.
- **Tryptophan (Trp, W):** Indole ring. Highest UV absorbance. Destroyed by acid hydrolysis [REF Q29].



Phenylalanine
(Phe, F)
F



Tyrosine
(Tyr)
Y



Tryptophan
(Trp)
W

Special Case:

- **Proline (Pro, P):** Cyclic imino acid (secondary amine). Rigid structure disrupts α -helices.



Proline

Exam Note:
REF Q16-2:
Phenylalanine
single letter
code is F.

Classification IV: Sulfur & Hydroxyl Groups

Sulfur-Containing:

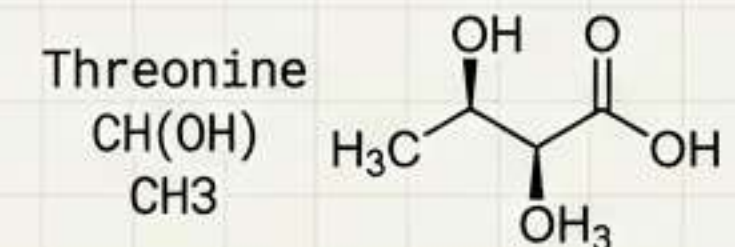
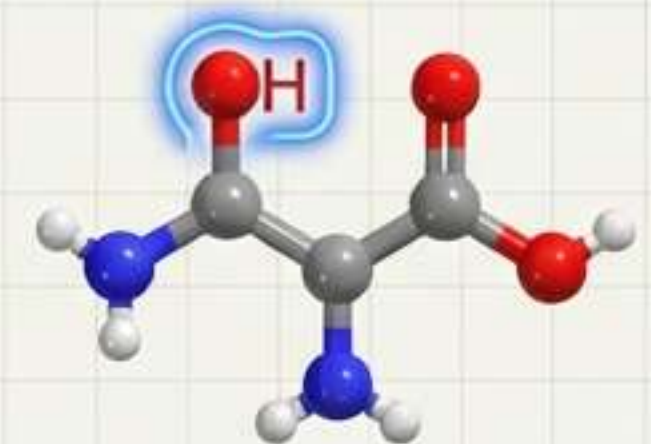
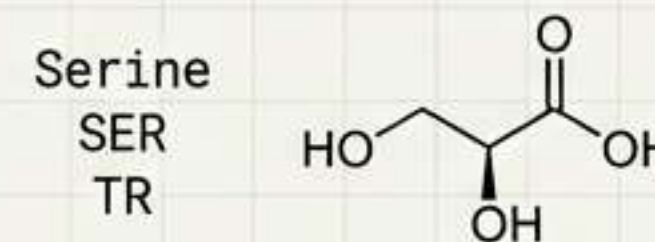
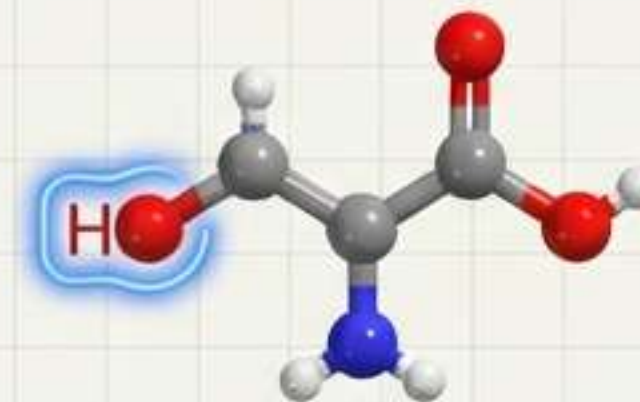
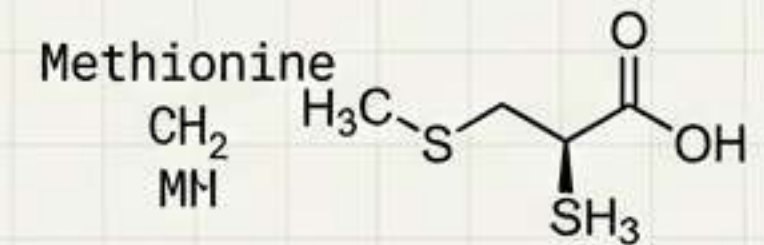
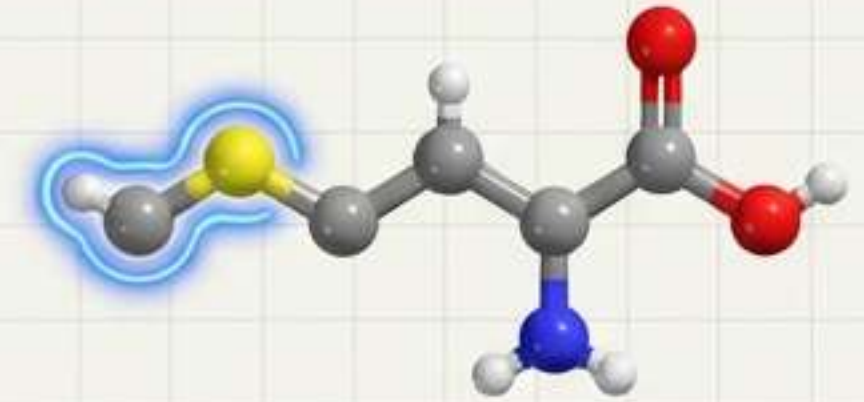
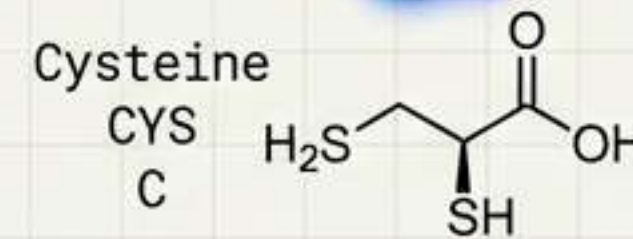
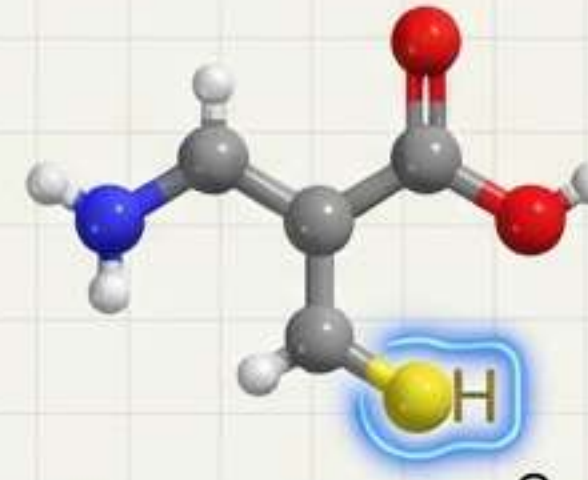
- **Cysteine (Cys, C):** Thiol (-SH). Forms covalent **Disulfide Bridges**.
- **Methionine (Met, M):** Thioether. Start codon (AUG). Cleaved by BrCN [REF Q42].

Hydroxyl-Containing (Alcohols):

- **Serine (Ser, S) & Threonine (Thr, T):** Polar uncharged.
- **Function:** Site of O-linked glycosylation and **Phosphorylation** (along with Tyr).

Exam Note:

REF Q5: Phosphorylation occurs on -OH groups (Ser, Thr, Tyr).



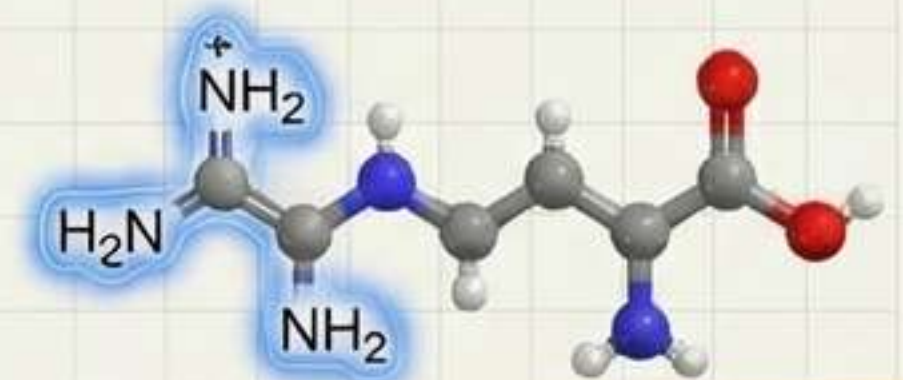
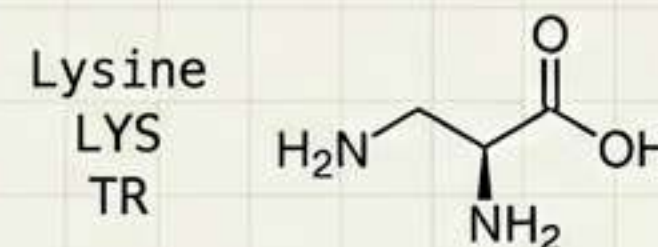
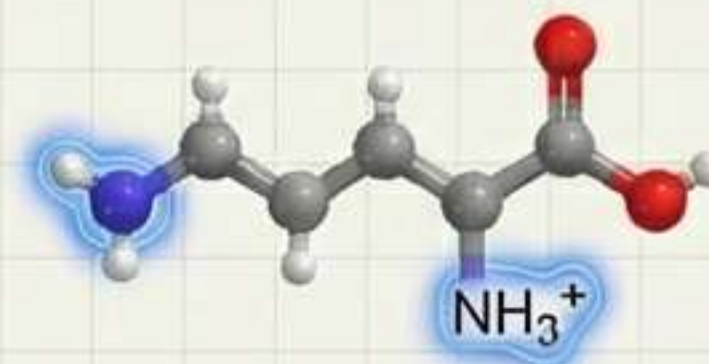
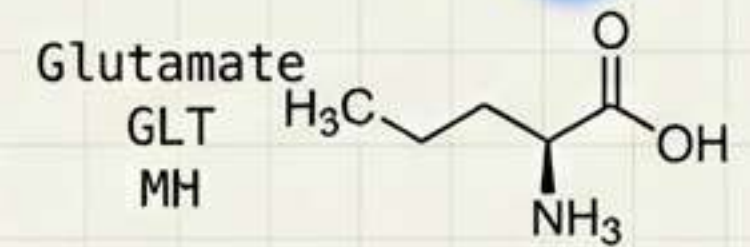
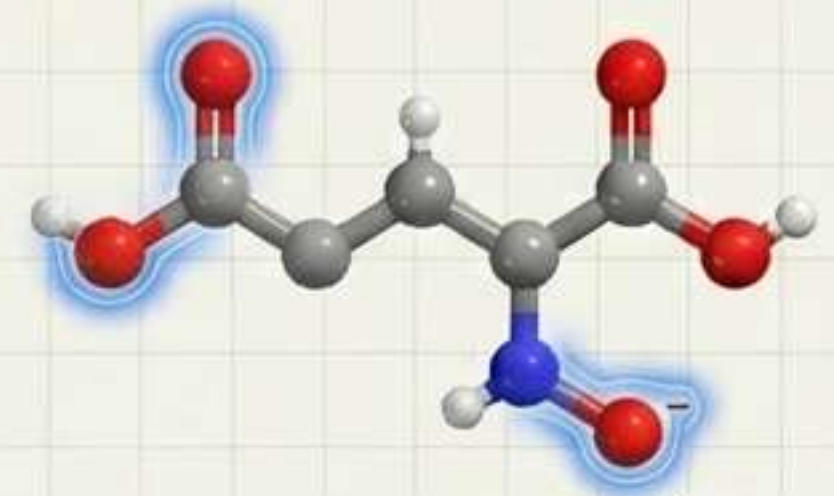
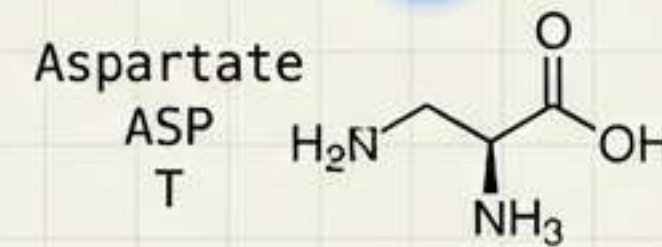
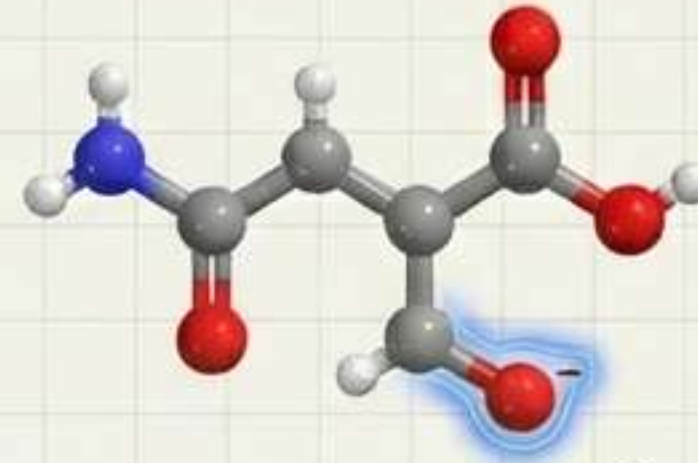
Classification V: The Charged Amino Acids

Acidic (Negatively Charged at pH 7):

- **Aspartate (Asp, D) & Glutamate (Glu, E):** Second carboxyl group.
- **Amide Derivatives:** Asparagine (Asn) and Glutamine (Gln).

Basic (Positively Charged at pH 7):

- **Lysine (Lys, K):** Primary amine at ϵ -carbon.
- **Arginine (Arg, R):** Guanidinium group. Very high pHi (~10.76).
- **Histidine (His, H):** Imidazole ring. Buffering capacity near neutral pH.



Exam Note:
REF Q16-4:
Aspartate
migrates to Anode
(+) at pH 6.3.

Nutritional Classification: The Essentials

Essential Amino Acids: Cannot be synthesized de novo. Must be ingested.

The List (8): Leu, Thr, Lys, Trp, Phe, Val, Met, Ile.

Mnemonic (French):

« *Le Très Lyrique Tristan Fait Vachement Méditer Iseult* »

(Leu - Thr - Lys - Trp - Phe - Val - Met - Ile)

Semi-Essential: Arginine & Histidine (Growth/Infants).

Exam Note:
REF Q23: Valine is Essential.

LEUCINE
Roboto Mono — THREONINE
LYSINE — Roboto Mono
Roboto Mono — TRYPTOPHAN
PHENYLALANINE
VALINE — Roboto Mono
METHIONINE
ISOLEUCINE

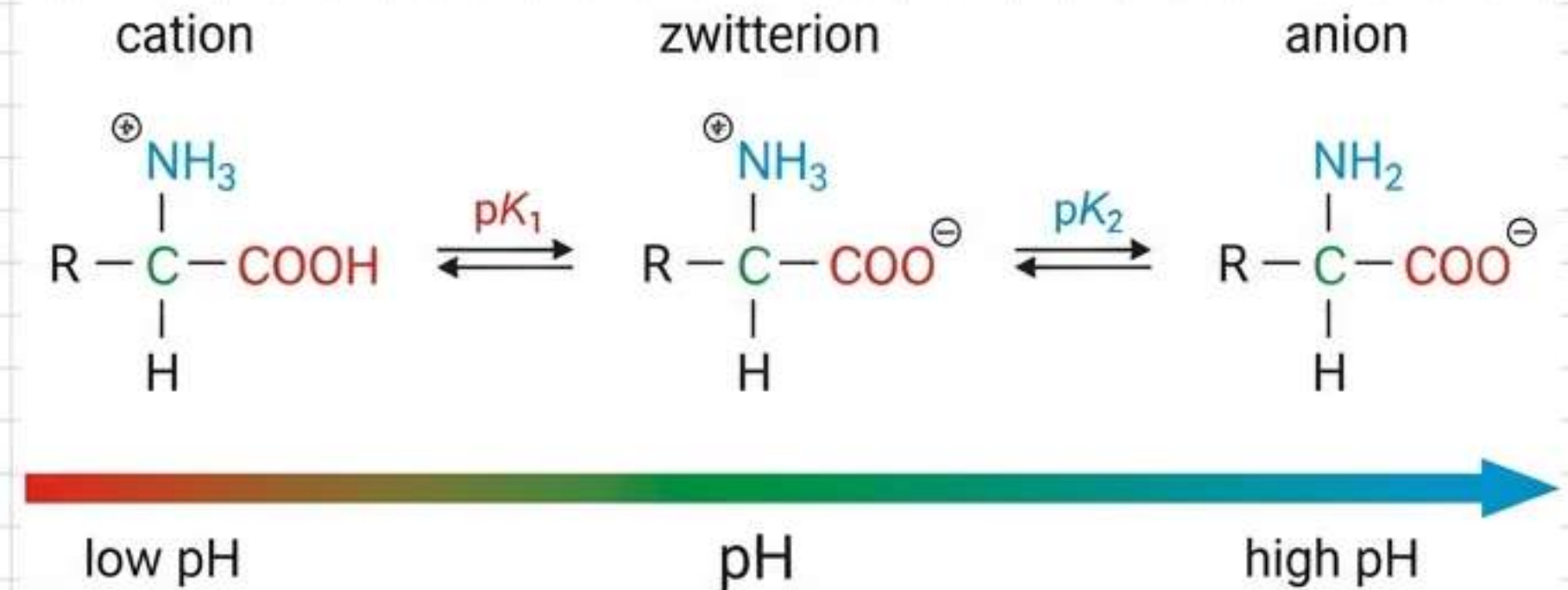
Physical Properties: Ionization & Zwitterions

Amphoteric Nature: Acts as both acid (proton donor) and base (proton acceptor).

The Zwitterion:

- * At $\text{pH} = \text{pH}_i$, the molecule is electrically neutral (Net charge = 0).
- * Maximal number of charges, but net is zero.
- * **Properties at pH_i :** Minimal solubility (precipitation risk), No migration in electric field.

Exam Note:
REF Q30: At pH_i , the compound does not migrate.



The Isoelectric Point (pHi): Calculations

Definition: The pH at which net charge is zero.

Calculation Rules:

1. **Neutral AA:** Average of pKa1 (COOH) and pKa2 (NH₃).

2. **Acidic AA (Asp, Glu):** Average of the two *lowest* pKas (COOH groups).

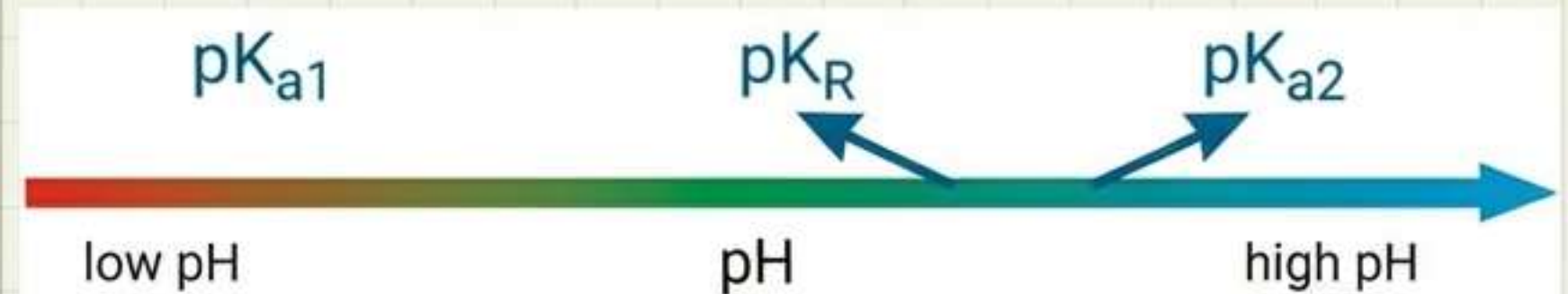
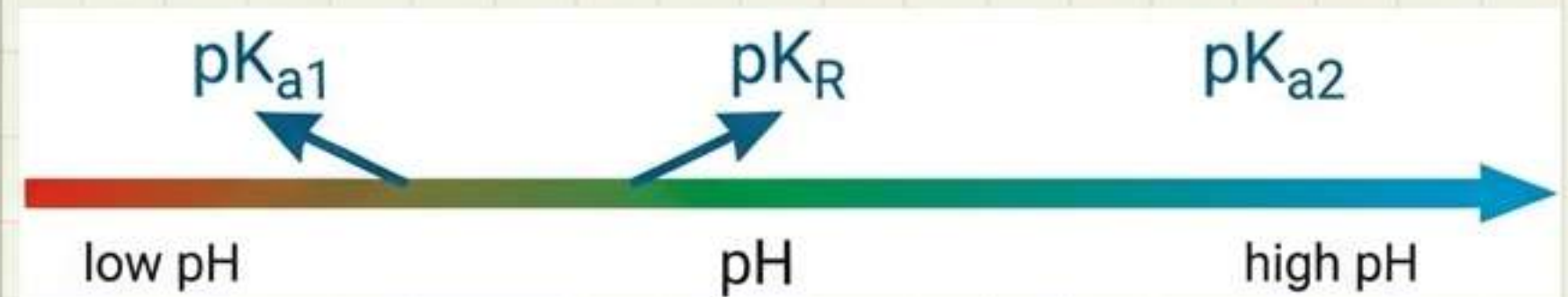
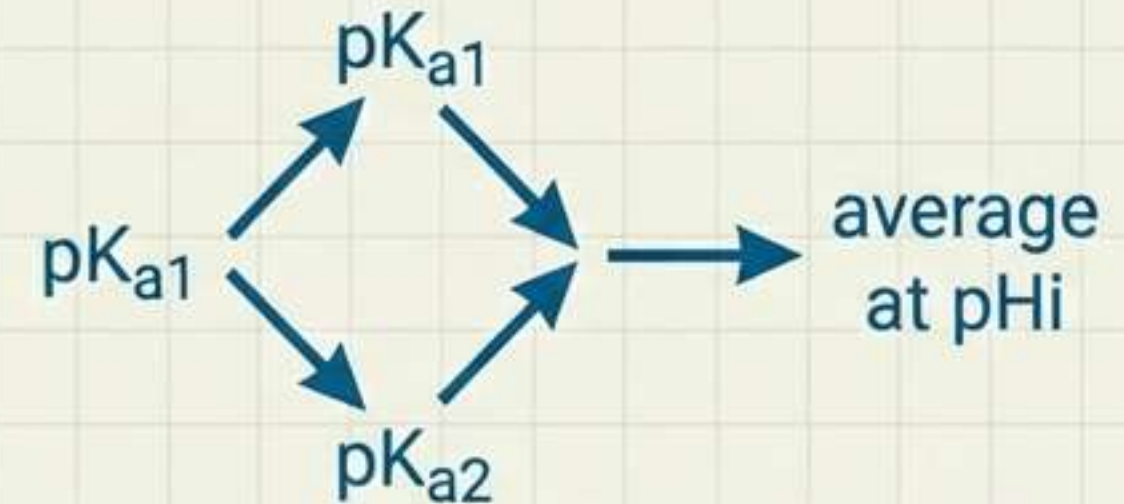
$$pHi = (pKa_1 + pK_R) / 2$$

3. **Basic AA (Lys, Arg, His):** Average of the two *highest* pKas (Amino groups).

$$pHi = (pK_R + pKa_2) / 2$$

Exam Note:

REF Q24:
Glutamate
(2.1, 9.5, 4.1).
 $pHi = (2.1 + 4.1) / 2 = 3.1$



Protein Structure: Primary & Secondary

Primary Structure:

- Linear sequence of amino acids.
- Linkage: **Peptide Bond** (Covalent, Amide, Rigid/Planar).

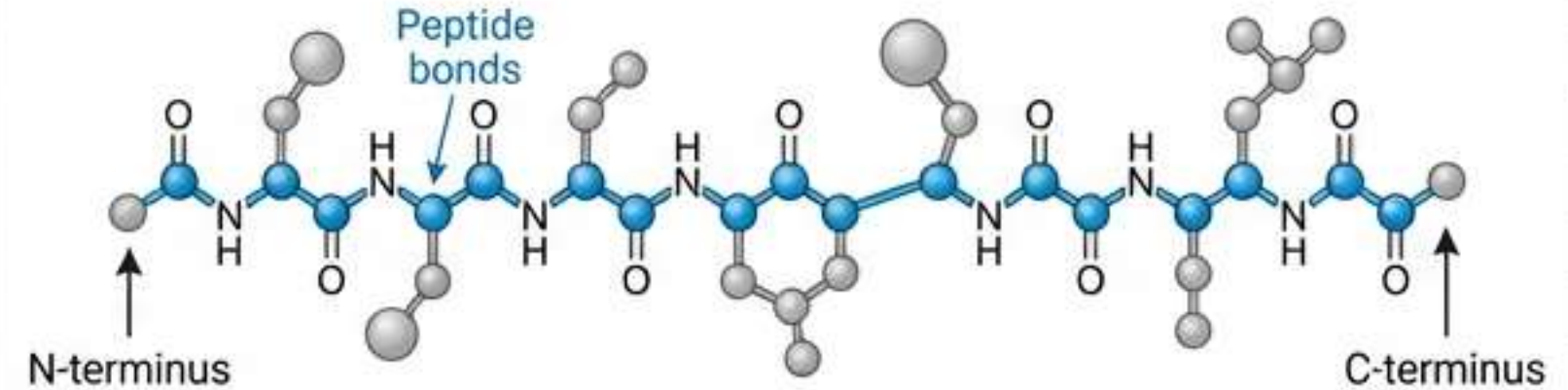
Secondary Structure:

- **α -Helix**: Right-handed spiral. 3.6 residues/turn. Stabilized by **H-bonds** (n to n+4) parallel to axis. [REF Q11]
- **β -Sheet**: Parallel or Anti-parallel strands. H-bonds perpendicular to axis.
- **β -Turns**: Reverses direction. Often contains Pro & Gly.

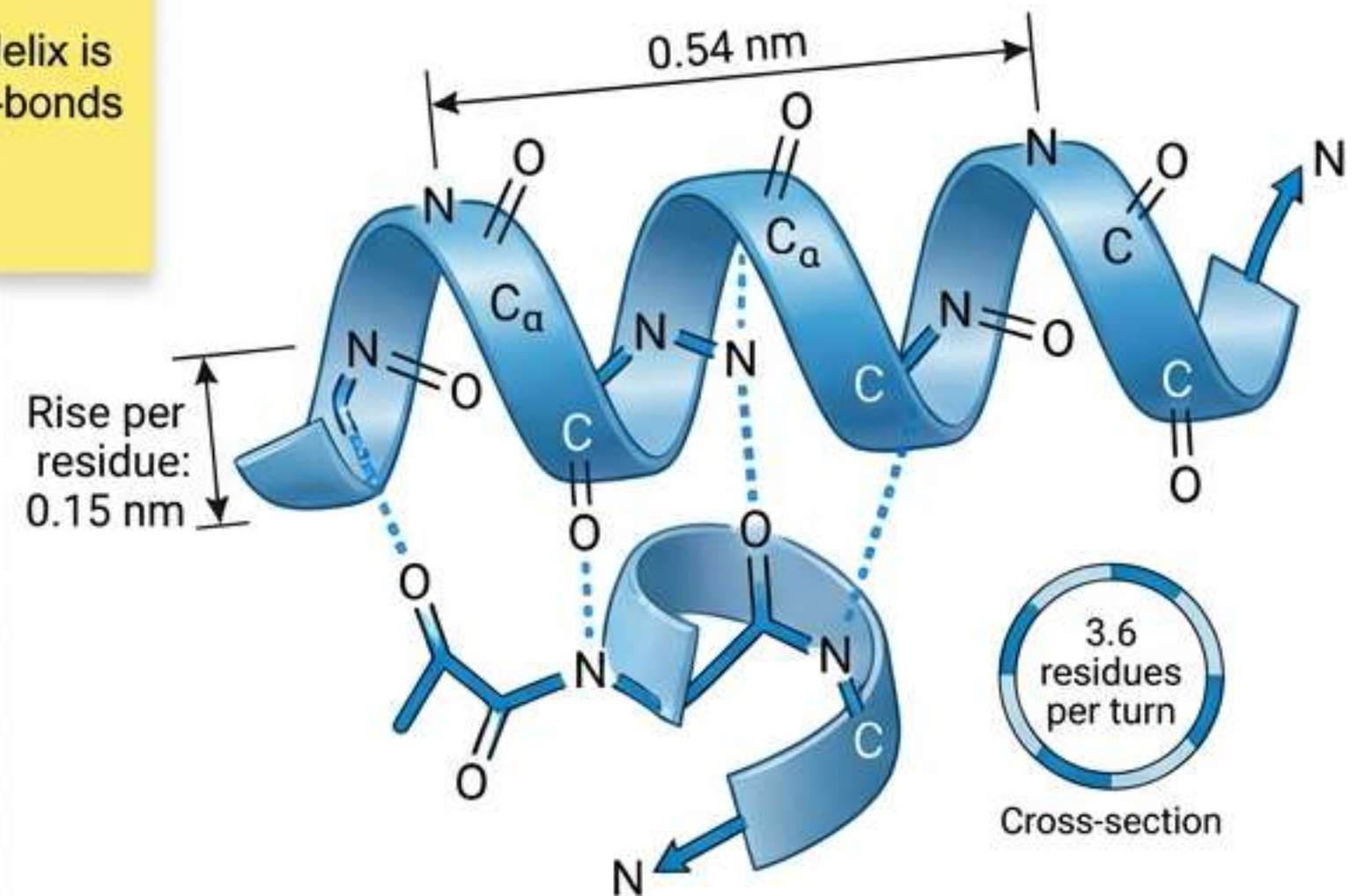
Exam Question

[REF Q11]: α -Helix is stabilized by H-bonds parallel to axis.

Primary Structure



Secondary Structure: α -Helix



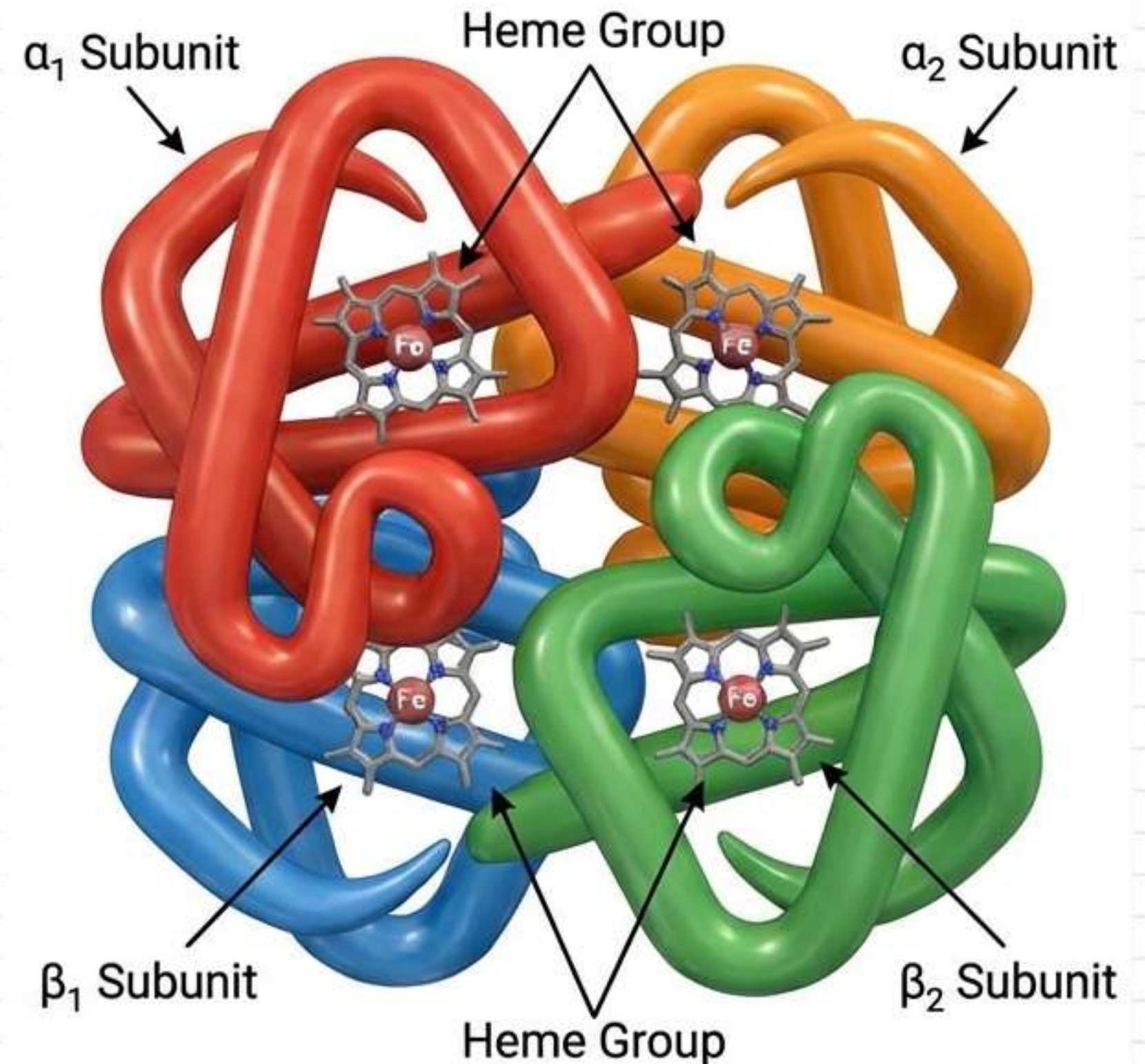
Tertiary & Quaternary Structure

Tertiary Structure:

- 3D folding of a single chain. Essential for function [REF Q8]. (in Roboto Mono)
- Stabilization:
 - **Hydrophobic:** Non-polar side chains cluster inside (core).
 - **Ionic:** Salt bridges between Acid/Base groups [REF Q39].
 - **Covalent:** Disulfide bridges (Cys-Cys).

Quaternary Structure:

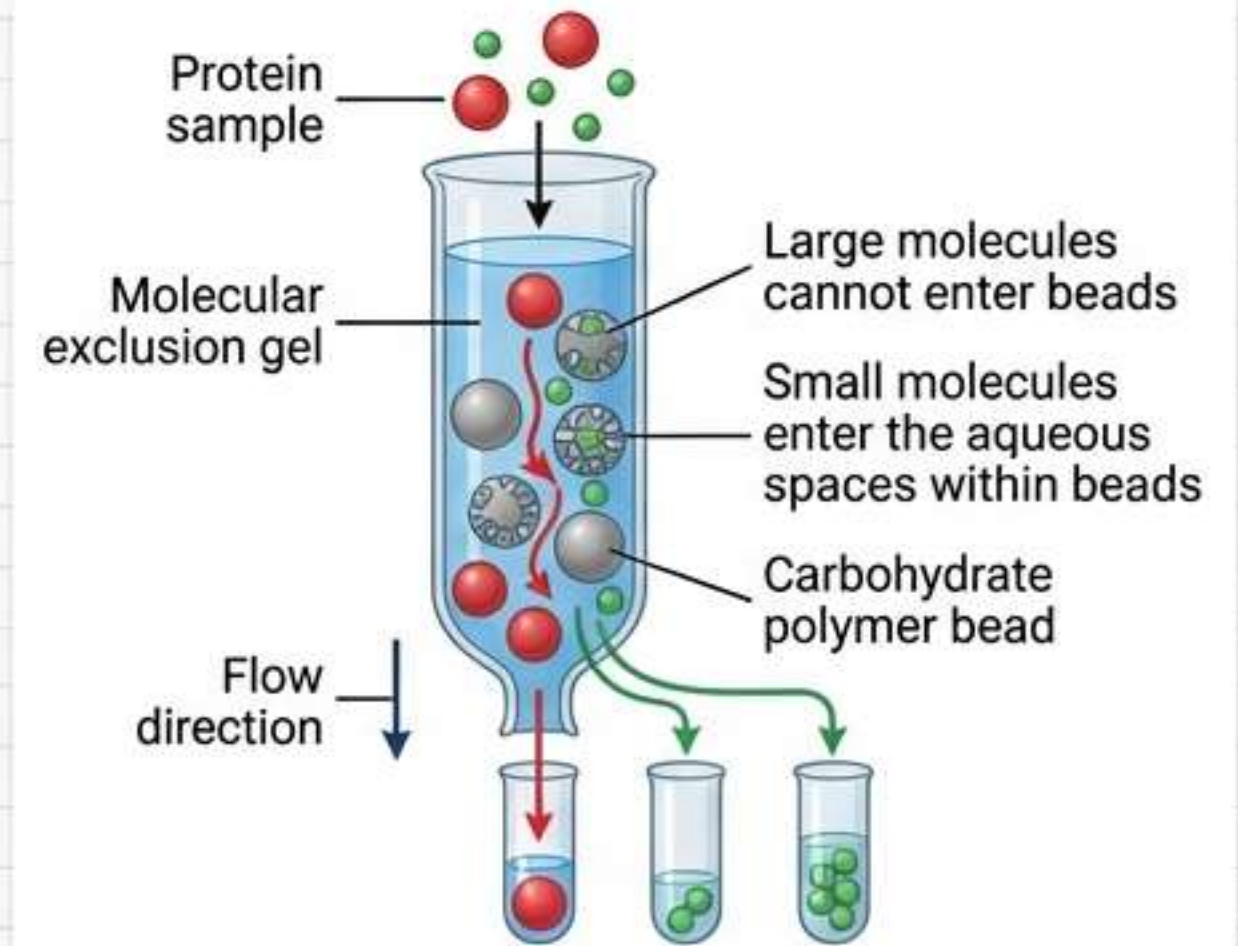
- Assembly of multiple subunits (protomers).
- **Example:** Hemoglobin ($\alpha_2\beta_2$). A Heteroprotein and Chromoprotein [REF Q34].



Methods: Chromatography Principles

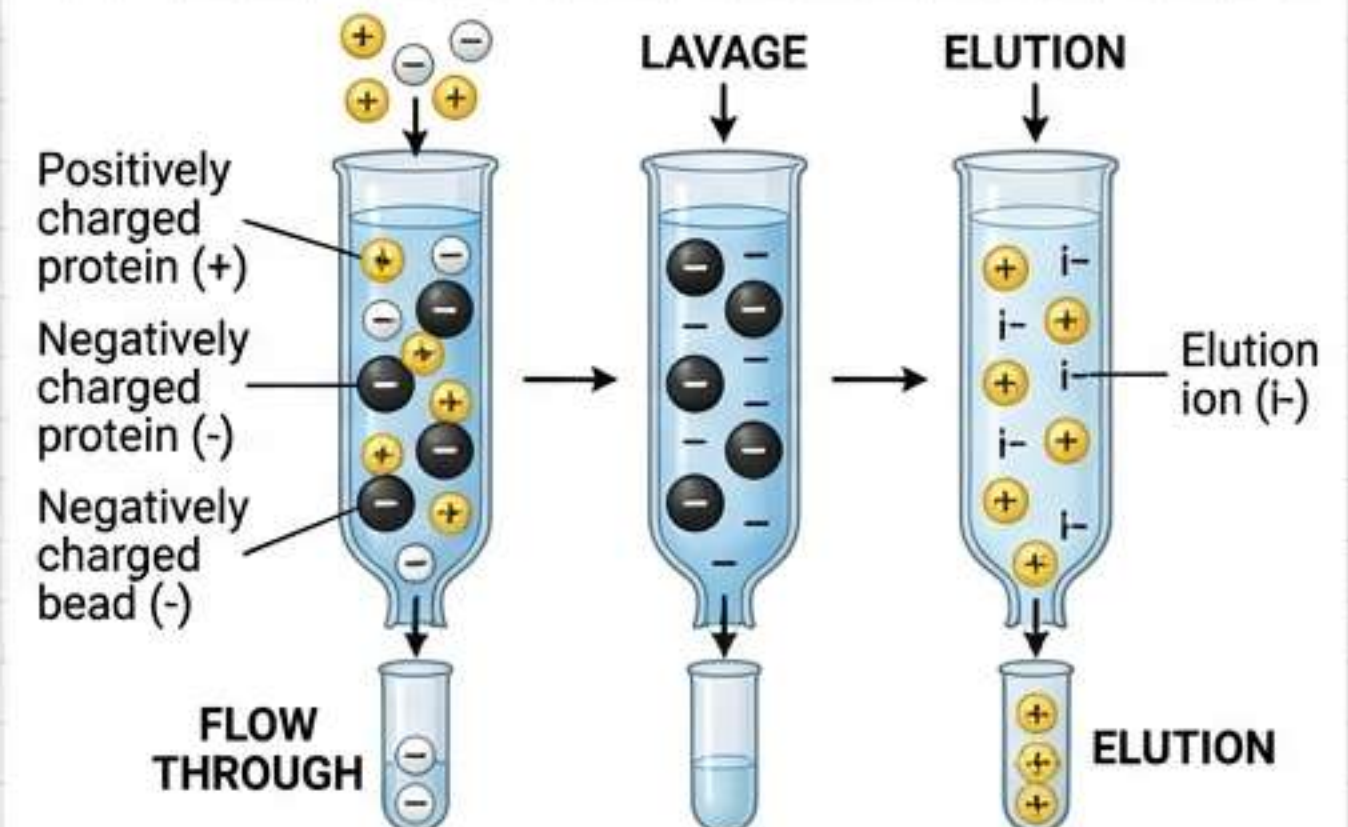
1. Gel Filtration (Exclusion):

- Separation by **SIZE**. Large molecules elute **FIRST** (cannot enter beads). Small molecules elute last.



2. Ion Exchange:

- Separation by **CHARGE**.
- **Cation Exchange (Neg beads):** Binds (+) proteins. Elute by increasing pH.
- **Order of Elution:** Lowest pI (Acidic) → Highest pI (Basic).
- *Example:* Asp → Leu → Lys [REF Q25].



Methods: Electrophoresis & SDS-PAGE

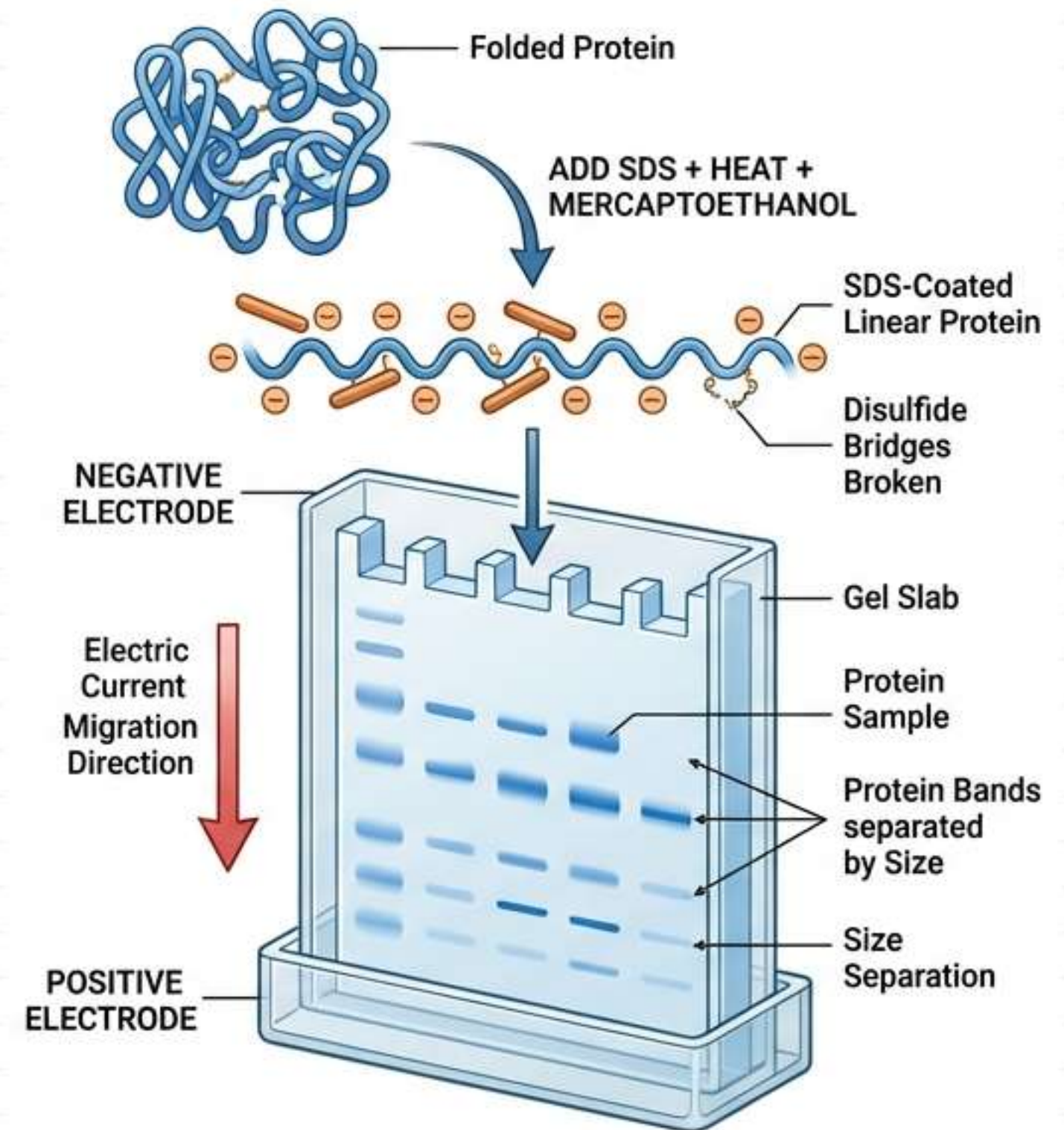
1. Zone Electrophoresis:

- Separates by Charge & Size.
- **Serum Profile:** Albumin (fastest) → γ -globulins (slowest).

2. SDS-PAGE (Denaturing):

- **SDS:** Anionic detergent unfolds protein + confers uniform negative charge.
- **Mechanism:** Separation strictly by **MASS (Molecular Weight)**.
- **Migration:** Small proteins migrate **faster**.
- **Note:** Mercaptoethanol is added to break disulfide bridges [REF Q31].

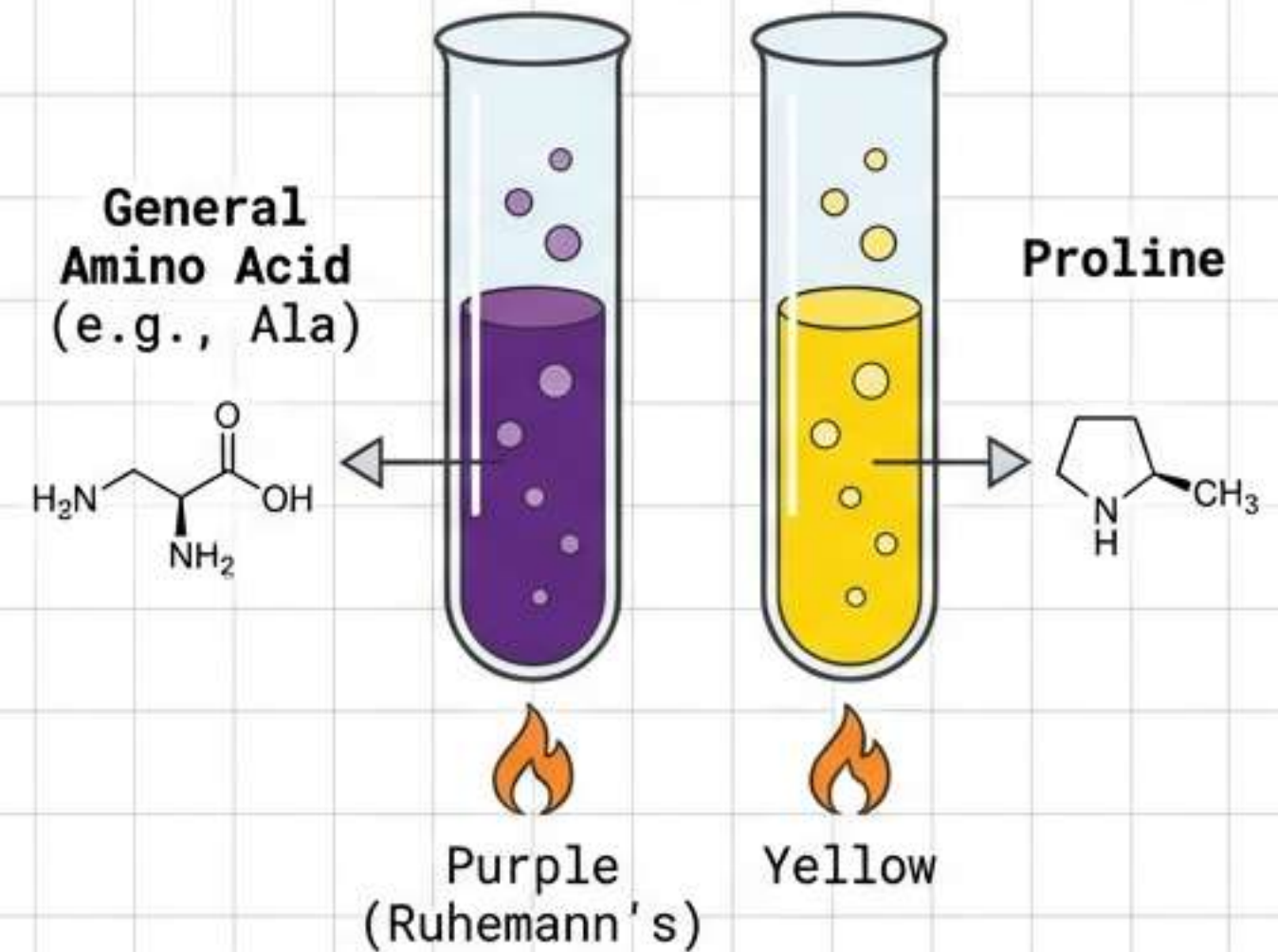
Exam Questions:
REF Q15/41: SDS-PAGE separates by mass, not charge.



Reactions & Identification

Detection:

- **Ninhydrin:** Reacts with α -NH₂.
 - **Result:** Purple (Ruhemann's purple).
 - **Exception: Proline** (secondary amine) turns **YELLOW** [REF Q16-1].
- **Biuret:** Detects peptide bonds (Violet).



Sequencing:

- **Edman Degradation:** N-terminal sequencing.
- **Cyanogen Bromide (BrCN):** Cleaves after **Methionine**.

REF Q29:
Acid hydrolysis
destroys
Tryptophan.

EXAM STRATEGY: Common Traps

Trap 1: Chirality

Question: "All amino acids are chiral."

Answer: FALSE. **Glycine** is **achiral**.

Trap 2: Migration Direction

Rule: If $\text{pH} < \text{pHi}$ → Protein is (+), migrates to Cathode (-).

Rule: If $\text{pH} > \text{pHi}$ → Protein is (-), migrates to Anode (+).

Trap 3: Hydrophobic Interactions

Question: "Hydrophobic bonds form between polar groups."

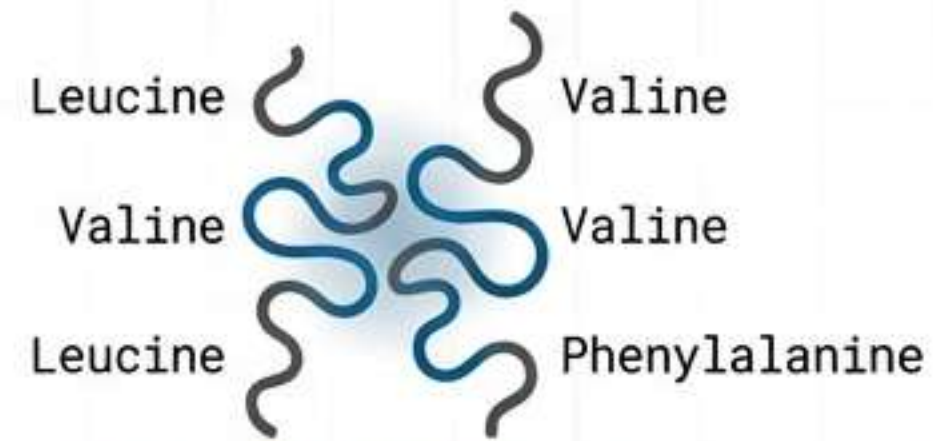
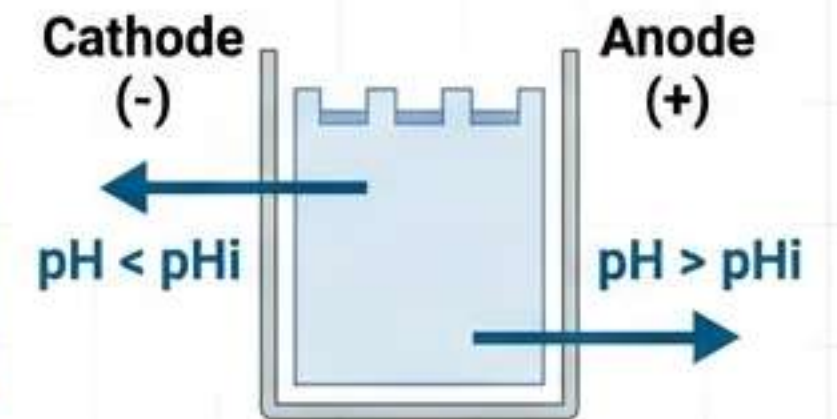
Answer: FALSE. Only between **non-polar** side chains.



Professor's Strategy



Exam Questions:
Be alert for these specific exceptions and rules! They are frequent pitfalls.



Hydrophobic Interaction
(Non-polar side chains ONLY)



No Hydrophobic Bonds
(Polar groups)

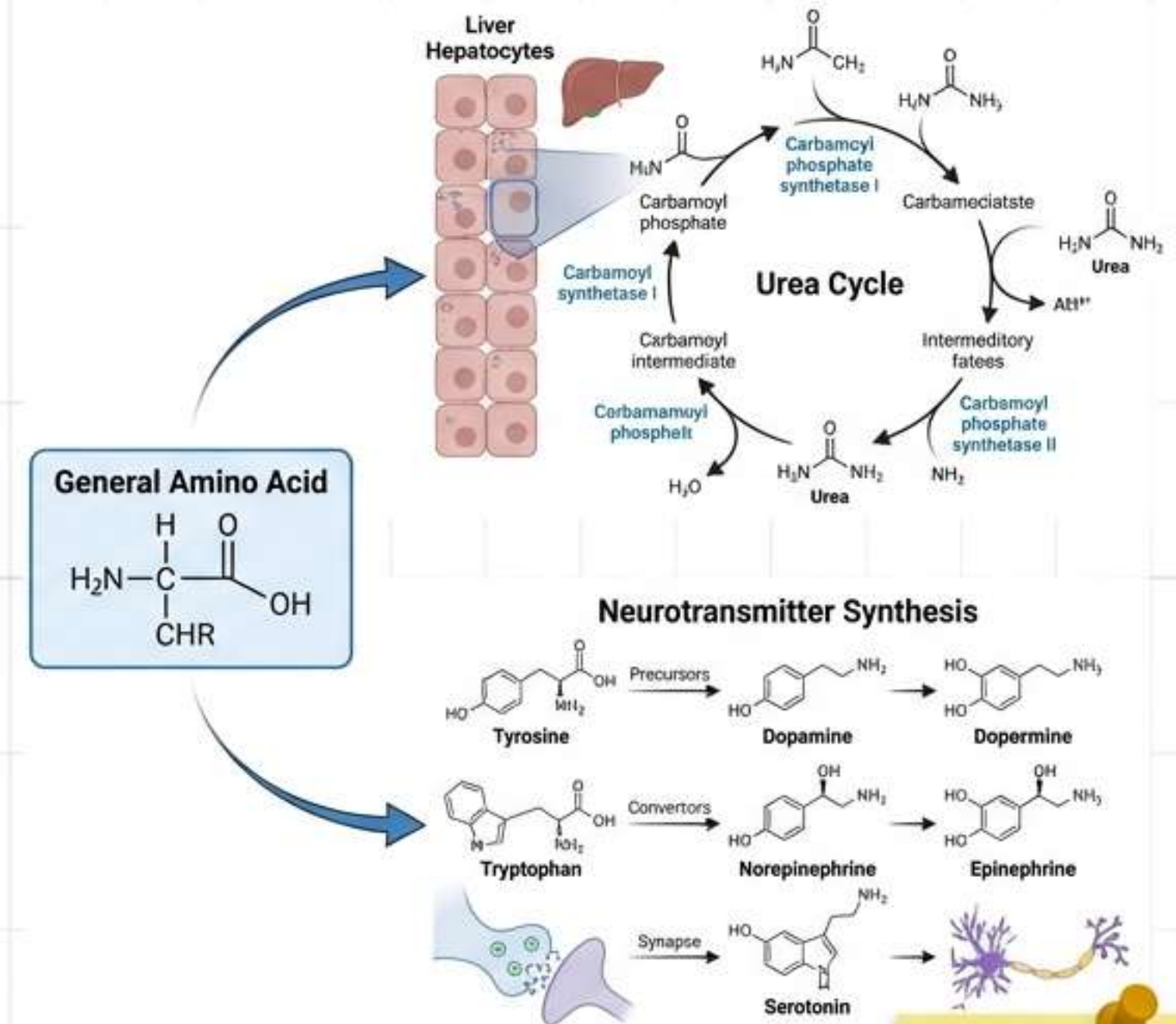
Clinical Relevance & Summary

Why this matters:

- * **Pathology:** Proteinuria (Kidney damage), Monoclonal Gammopathy (Spike in γ -globulins).
- * **Metabolism:** Urea Cycle eliminates Nitrogen. Ketogenic/Glucogenic fates.

Final Review Checklist:

1. Memorize the **Non-Polar** list.
2. Master the **pHi Calculation** for Acidic/Basic AAs.
3. Know the **Separation Principles** (Charge vs Size).



EXAM QUESTIONS:
Be alert for these specific exceptions and rules! They are frequent pitfalls.